Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2024-P1-Model)** **Date: 03-05-2026**

Time: 09.00Am to 12.00Pm **GTA-29** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**A**	2	**C**	3	**A**	4	**B**	5	**ACD**	6	**C**
7	**CD**	8	**3**	9	**800**	10	**143**	11	**23**	12	**2025**
13	**78**	14	**B**	15	**C**	16	**B**	17	**A**		

PHYSICS

18	**A**	19	**B**	20	**C**	21	**C**	22	**ACD**	23	**ACD**
24	**ABC**	25	**4**	26	**3**	27	**5**	28	**5**	29	**354**
30	**48**	31	**A**	32	**A**	33	**B**	34	**D**		

CHEMISTRY

35	**B**	36	**B**	37	**A**	38	**D**	39	**ACD**	40	**BCD**
41	**ACD**	42	**6**	43	**11**	44	**35**	45	**5**	46	**2**
47	**146**	48	**A**	49	**A**	50	**A**	51	**D**		

For More Join: @IITJEE_Advanced

SOLUTIONS
MATHEMATICS

01. Let $t=\tan x>0$ and define $\theta=\tan^{-1}(at)\in\left(0,\frac{\pi}{2}\right)$. Then using the identify

$\sin(2\pi)=\frac{2\tan\theta}{1+\tan^2\theta}$,

We get $\frac{2a\tan a}{1+a^2\tan^2 x}=\sin(2\theta)$.

So the equation becomes $\sin^{-1}(\sin 2\theta)=\frac{\pi}{2}-x$.

Now $2\theta\in(0,\pi)$, hence $\sin^{-1}(\sin 2\theta)=\begin{cases}2\theta, & 0<2\theta\le\frac{\pi}{2}\Leftrightarrow at\le 1,\\ \pi-20, & \frac{\pi}{2}<2\theta<\pi\Leftrightarrow at>1.\end{cases}$

Case 1: $at\le 1$. Then $2\theta=\frac{\pi}{2}-x$. Taking tangent:

$\tan\left(\frac{\pi}{2}-x\right)=\cot x=\frac{1}{t}$.

Also $\tan(2\theta)=\tan(2\tan^{-1}(at))=\frac{2at}{1-a^2t^2}$.

So $\frac{1}{t}=\frac{2at}{1-a^2t^2}\Rightarrow 1-a^2t^2=2at^2\Rightarrow t^2=\frac{1}{a(a+2)}$.

This gives exactly one t > 0 for every $a>0$. Moreover, $at=\sqrt{\frac{a}{a+2}}<1$,

So it indeed lies in Case 1.

Case 2: $at>1$. Then $\pi-2\theta=\frac{\pi}{2}-x\Rightarrow 2\theta=\frac{\pi}{2}+x$. Taking tangent:

$\tan\left(\frac{\pi}{2}+x\right)=-\cot x=-\frac{1}{t}$.

So $\frac{2at}{1-a^2t^2}=-\frac{1}{t}\Rightarrow 2at^2=a^2t^2-1\Rightarrow t^2=\frac{-1}{a(2-a)}$.

For $t^2>0$ with $a>0$, we must have $a>2$. Then we get exactly one more solution

$t^2=\frac{1}{a(a--2)}$, And indeed $at=\sqrt{\frac{a}{a-2}}>1$,

So this solution is valid in Case 2.

Hence $N(a)=\begin{cases}1, & 0<a\le 2,\\ 2, & a>2.\end{cases}$

Answer: $N(a)=1$ for $0<a\le 2$, and $N(a)=2$ for $a>2$. Option: (A).

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02. A tangent to $y^2 = 4x$ with parameter t is $ty = x + t^2$,

Touching the parabola at $(t^2, 2t)$. If tangent from P (h, k) correspond to $t_1 \neq t_2$, then $t_1k = h + t_1^2$, $t_2k = h + t_2^2$.

Subtracting gives $k = t_1 + t_2$.

Then $h = t_1k - t_1^2 = t_1(t_1 + t_2) - t_1^2 = t_1t_2$.

Let $s = t_1 + t_2$, $p = t_1t_2$.

Then (h, k) = (p, s) and the circle condition gives $p^2 + s^2 = 5$ .....(1)

Also, $(t_2 - t_1) = (t_1 + t_2)^2 - 4t_1t_2 = s^2 - 4p$ .....(2)

Step (1): Express area in terms of $(t_2 - t_1)$.

The required area (between the parabola and the two tangents from the contact points) evaluates to $S(P) = \dfrac{(t_2 - t_1)^3}{6}$ .....(3)

(direct integration along y from $2t_1$ to $2t_2$ gives this standard result for $y^2 = 4x$). So maximizing S(P) is equivalent to maximum $(t_2 - t_1)^2$.

Step (2): Maximize $(t_2 - t_1)^2 = s^2 - 4p$ using only (1) and basic inequality.

From (1), $s^2 = 5 - p^2$. Substituting into (2): $(t_2 - t_1)^2 = (5 - p^2) - 4p = 9 - (p + 2)^2$.

Since $(p + 2)^2 \geq 0$, we get $(t_2 - t_1)^2 \leq 9$,

With equality when p = –2. Thus the maximum is $(t_2 - t_1)^2_{\max} = 9 \Rightarrow t_2 - t_1 = 3$.

Step (3): $\max S(P) = \dfrac{9}{2}$. Option: (C).

03. Since urn choice is uniform, $\mathbb{P}(U_1|E) = \dfrac{\mathbb{P}(E|U_1)}{\mathbb{P}(E|U_1) + \mathbb{P}(E|U_2) + \mathbb{P}(E|U_3)}$.

Let the chosen urn have r red and b black with r + b = 4.

Route A: first two draws are RR, $\mathbb{P}(RR) = \dfrac{r}{4}.\dfrac{r-1}{3} = \dfrac{r(r-1)}{12}$.

The draw one more from remaining two balls. To make total reds = 3, the third draw must be red: $\mathbb{P}(3\text{rd is } R|RR) = \dfrac{r-2}{2}$.

So $\mathbb{P}(\text{E via } RR) = \dfrac{r(r-1)}{122}.\dfrac{r-2}{2} = \dfrac{r(r-1)(r-2)}{24}$.

(If first two are BB, we cannot reach 3 reds).

Route B: first two draws are different (RB or BR).

$\mathbb{P}(\text{different}) = \dfrac{r}{4}.\dfrac{b}{3} + \dfrac{b}{4}.\dfrac{r}{3} = \dfrac{rb}{6}$

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These two draws already contain exactly and red. After returning them, we draw 3 balls from the original urn. We now exactly two reds in the last 3 draws, and the last (5th) draw must be red.

In 3 draws, with last red and exactly 2 reds total, the only patterns are RBR and BRR:

$\mathbb{P}(RBR)=\frac{r}{4}.\frac{b}{3}.\frac{r-1}{2}=\frac{rb(r-1)}{24}$.

$\mathbb{P}(\text{BRR})=\frac{b}{4}.\frac{r}{3}.\frac{r-1}{2}=\frac{rb(r-1)}{24}$.

So $\mathbb{P}(\text{last 3 give 2R and last R})=\frac{rb(r-1)}{12}$.

Hence $\mathbb{P}(\text{E via different})=\frac{rb}{6}.\frac{rb(r-1)}{12}=\frac{r^2b^2(r-1)}{72}$.

Therefore $\mathbb{P}\left(E\middle|r,b\right)=\frac{r(r-1)(r-2)}{24}+\frac{r^2b^2(r-1)}{72}$.

Now compute:

For $U_1:(r,b)=(3,1):\mathbb{P}\left(E\middle|U_1\right)=\frac{3.2.1}{24}+\frac{9.1.2}{72}=\frac{1}{4}+\frac{1}{4}=\frac{1}{2}$.

For $U_2:(r,b)=(2,2):\mathbb{P}\left(E\middle|U_2\right)=0+\frac{4.4.1}{72}=\frac{16}{72}=\frac{2}{9}$.

For $U_3:(r,b)=(1,3):$ impossible to observe 3 reds, so $\mathbb{P}\left(E\middle|U_3\right)=0$.

Thus $\mathbb{P}\left(U_1\middle|E\right)=\frac{\frac{1}{2}}{\frac{1}{2}+\frac{2}{9}}=\frac{\frac{1}{2}}{\frac{13}{18}}=\frac{9}{13}$.

04. $f_a(-1)=-1+3+a=2+a$, $f_a(2)=8-6+2a=2+2a$.

So slope of chord ℓ: $m=\frac{(2+2a)-(2+a)}{3}=\frac{a}{3}$.

Derivative (where it exists): $f_a'(x)=\begin{cases}3x^2-3-a, & x<0,\\ 3x^2-3+a, & x>0.\end{cases}$.

At x = 0, left derivative = –3 –a and right derivative = –3 + a, so tangent exists at 0 iff a = 0. But we need $c\in(-1,2)$, and we solve separately in (–1, 0) and (0, 2).

We need $f_a'(c)=\frac{a}{3}$.

i) For $c\in(0,2)$: $3c^2-3+a=\frac{a}{3}\Rightarrow 9c^29+2a=0\Rightarrow c^2=1-\frac{2a}{9}$.

A solution exists in (0, 2) iff $0<c^2<4$, i.e. $1-\frac{2a}{9}>0\Rightarrow a<\frac{9}{2},1-\frac{2a}{9}<\Leftrightarrow a>-\frac{27}{2}$.

Hence this contributes exactly one c when $-\frac{27}{2}<a<\frac{9}{2}$.

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ii) for $c \in (-1,0)$: $3c^2 - 3 - a = \frac{a}{3} \Rightarrow 9c^2 - 9 - 4a = 0 \Rightarrow c^2 = 1 + \frac{4a}{9}$.

A solution exists in (–1, 0) iff $0 < c^2 < 1$: $1 + \frac{4a}{9} > 0 \Rightarrow a > -\frac{9}{4}, 1 + \frac{4a}{9} < 1 \Rightarrow a < 0$.

So this contributes exactly one c when $-\frac{9}{4} < a < 0$.

Counting N(a):

- If $-\frac{27}{2} < a \le -\frac{9}{4}$: only (i) gives a solution $\Rightarrow N(a) = 1$.
- If $-\frac{9}{4} < a < 0$: both (i) and (ii) gives solutions $\Rightarrow N(a) = 2$.
- If $0 \le a < \frac{9}{2}$: only (i) gives a solution $\Rightarrow N(a) = 1$.
- Otherwise $N(a) = 0$.

Therefore $N(a) = 1 \Leftrightarrow a \in \left(-\frac{27}{2}, \frac{-9}{4}\right] \cup \left[0, \frac{9}{2}\right)$.

05. Write $1 + \frac{ia}{n} = r_n e^{i\alpha n}, \alpha_n = \tan^{-1}\left(\frac{a}{n}\right) \in \left(0, \frac{\pi}{2}\right)$.

Then $z_n(a) = r_n^n e^{in\alpha n}$.

So the (unreduced) argument is $u_n(a) = n\alpha_n = n\tan^{-1}\left(\frac{a}{n}\right)$.

The principal argument is

$\theta_n(a) = Atg\left(z_n(a)\right) =$ the unique value in $(-\pi, \pi]$ congruent to $u_n(a) (\mathrm{mod}\, 2\pi)$.

(D) is true. Fix a > 0 and consider the real function for x > 0: $f(x) = x\tan^{-1}\left(\frac{a}{x}\right)$.

Then $u_n(a) = f(n)$. Differentiate (standard 12th calculus):

$$f'(x) = \tan^{-1}\left(\frac{a}{x}\right) + x.\frac{1}{1+(a/x)^2}.\left(-\frac{a}{x^2}\right) = \tan^{-1}\left(\frac{a}{x}\right) - \frac{ax}{x^2+a^2}.$$

Let $t = \frac{a}{x} > 0$. Then $f'(x) = \tan^{-1}(t) - \frac{t}{1+t^2}$.

Now definite $g(t) = \tan^{-1}(t) - \frac{t}{1+t^2}$ for $t > 0$. Then

$$g'(t) = \frac{1}{1+t^2} - \frac{(1+t^2)-2t^2}{(1+t^2)^2} = \frac{1}{1+t^2} - \frac{1-t^2}{(1+t^2)^2} = \frac{2t^2}{(1+t^2)^2} > 0.$$

Also $g(0) = 0$, so $g(t) > 0$ for all t > 0. Hence $f'(x) > 0$ for all $x > 0$, so f is strictly increasing, therefore $u_n(a) = f(n)$ is strictly increasing in n.

Also, using $\tan^{-1}(y) \sim y$ as $y \to 0$ (standard limit), $\lim_{n\to\infty} u_n(a) = \lim_{n\to\infty} n\tan^{-1}\left(\frac{a}{n}\right) = a$.

So statement (D) holds.

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(A) is true and (B) is false. We have $u_n(a)$ strictly increasing and $u_n(a) \to a$.

- If $0 < a < \pi$, then $u_n(a) < a < \pi$ for all n, so no reduction modulo 2π occurs and $\theta_n(a) = u_n(a)$, hence $(\theta_n(a))$ is strictly increasing and $\lim \theta_n(a) = a \in (0,\pi)$. So $a \in S$.
- If $a = \pi$, then $\lim \theta_n(a) = \pi \notin (0,\pi)$, so $a \notin S$.
- If $\pi < a < 2\pi$, then fir large $n, u_n(a)$ becomes $> \pi$ (since it increase to a). Whenever $u_n(a) \in (\pi, 2\pi)$, the principal argument becomes $\theta_n(a) = u_n(a) - 2\pi \in (-\pi, 0)$,

So the sequence be strictly increasing across the "wrap" near π. Also

$\lim \theta_n(a) = a - 2\pi \in (-\pi, 0) \notin (0,\pi)$. So $a \notin S$.

Therefore $S = (0,\pi)$, (A) is true, and (B) is false.

(C) is true. We have $z_n(a) \in \mathbb{R}$ iff its argument is an integer multiple of π:

$$u_n(a) = n\tan^{-1}\left(\frac{a}{n}\right) = m\pi,\ m \in \mathbb{Z}.$$

Since $u_n(a) > 0$, we need $m \geq 1$. Also $\tan^{-1}(a/n) \in (0, \pi/2)$, so $\frac{m\pi}{n} < \frac{\pi}{2} \Rightarrow m < \frac{n}{2}$.

Thus for some $n \geq 3$, we can write $\tan^{-1}\left(\frac{a}{n}\right) = \frac{m\pi}{n} \Rightarrow \frac{a}{n} = \tan\left(\frac{m\pi}{n}\right) \Rightarrow a = n\tan\left(\frac{m\pi}{n}\right)$.

Now enforce $a \in (0, 2\pi)$. If $m \geq 2$, then $\frac{m\pi}{n} \geq \frac{2\pi}{n}$ and tan is increasing on $(0, \pi/2)$, so

$a = n\tan\left(\frac{m\pi}{n}\right) \geq n\tan\left(\frac{2\pi}{n}\right)$.

But for $0 < x < \frac{\pi}{2}$, $\tan x > x$, hence $n\tan\left(\frac{2\pi}{n}\right) > n.\frac{2\pi}{n} = 2\pi$, so $a > 2\pi$, impossible.

Therefore only m = 1 can work. Hence $R = \left\{n\tan\left(\frac{\pi}{n}\right) : n \in \mathbb{N}, n \geq 3\right\} \cap (0, 2\pi)$, which is exactly statement (C).

06. Volume $= \left|\left[2\vec{b}\times\vec{c}\ \ 3\vec{c}\times\vec{a}\ \ 4\vec{a}\times\vec{b}\right]\right| = 18$, $24\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = 18 \Rightarrow \left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right| = \frac{\sqrt{3}}{2}$

Now, $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} (1+\sin\theta) & \cos\theta & \sin 2\theta \\ \sin\left(\theta+\frac{2\pi}{3}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & \sin\left(2\theta+\frac{4\pi}{3}\right) \\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right) \end{vmatrix}$

Applying $R_1 \to R_1 + R_2 + R_3$ we get after simplification,

$$\left|\left[\vec{a}\vec{b}\vec{c}\right]\right| = \sqrt{3}\left|\cos 3\theta\right| = \frac{\sqrt{3}}{2} \Rightarrow \cos 3\theta = \pm\frac{1}{2} \Rightarrow 3\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3} \Rightarrow \theta = \frac{\pi}{9}, \frac{2\pi}{9}, \frac{4\pi}{9}.$$

07. A.M. of distances of focus from two directrices is CP i.e. 26

08. LHS is even and RHS is odd

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09. Take 2 cases: one for having 5,6 in between 7 and 8 and the other for having 5,6 not between 7,8. Number of ways will be 18200

10. $|Q-6I|=|P-6I|=\begin{vmatrix}-2 & 3 & 4\\ 3 & 0 & 5\\ 4 & 5 & 3\end{vmatrix}=143$

11.

Class	f	cf
0-10	8	8
10-20	30	38
20-30	40	78
30-40	12	90
40-50	10	100

Here $\frac{N}{2}=50$ which lies in the value 78 of cf. Hence corresponding class of this cf is 20-30 is the median class. So, $l=20, f=40, F=38, h=10 \Rightarrow \text{Median}=l+\frac{\frac{N}{2}-F}{f}h=23$

12. Numerator is $n\int_0^1 \{x\}\,dx = n\int_0^1 x\,dx$

13. $\det(A)=(y-x)(z-x)(z-y)$.

Let $d_1=y-x, d_2=z-y \Rightarrow z-x=d_1+d_2$.

Then $(y-x)(z-x)(z-y)=d_1(d_1+d_2)d_2=240$ (*).

Also, since $x,y,z\in\{0,....,16\}$, all differences satisfy $|d_1|,|d_2|,|d_1+d_2|\le 16$.

Step 1: Find integer pairs (d_1,d_2) satisfying (*) with bounds. First consider $d_1>0, d_2>0$. Then d_1, d_2, d_1+d_2 are positive integers ≤ 16 and multiply to 240.

The only such triples are: (1, 15, 16), (2, 10, 12), (4, 6, 10), giving ordered pairs $(d_1d_2)\in\{(1,15),(15,1),(2,10),(10,2),(4,6),(6,4)\}$.

Next, for product to remain positive with mixed signs, we need exactly two negatives among d_1, d_2, d_1+d_2. That yields the additional ordered pairs: (–16, 1), (–16, 15), (–12, 10), (–10, 4), (–10, 6) and their swaps with $d_1 \leftrightarrow d_2$: (1, –16), (15, –16), (2, –12), (10, –12), (4, –10), (6, –10). In total there are 18 ordered pairs (d_1,d_2).

Step 2: Fir each (d_1,d_2), count valid x.

We have $y=x+d_1$, $z=x+d_1+d_2$ and we need $x,y,z\in[0,16]\cap\mathbb{Z}$.

- If $(d_1,d_2)=(1,15)$ or (15, 1), then $z=x+16\le 16 \Rightarrow x=0$ (1 value). Similarly for (–16, 1), (1, –16), (–16, 15), (15, –16), we get exactly 1 value of x. So there are 6 such ordered pairs each contributing 1: Total 6.
- If $(d_1,d_2)=(2,10)$ or (10, 2), then $z=x+12\le 16 \Rightarrow x=0,1,2,3,4$ (5 values). Similarly for (–12, 2), (2, –12), (–12, 10), (10, –12) each contributes 5. So there are 6 such ordered pairs each contributing 5: Total 30.

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- If $(d_1,d_2)=(4,6)$ or (6, 4), then $z=x+10\le 16\Rightarrow x=0,1,.....,6$ (7 value). Similarly for (–10, 4), (4, –10), (–10, 6), (6, –10) each contributes 7. So there are 6 such ordered pairs each contributing 7: Total 42.

Hence N = 6 + 30 + 42 = 78.

14. P(drawn in 1 round) $=\frac{6}{36}=\frac{1}{6}$, P(win in 1 round) $=\frac{1}{2}\left(1-\frac{1}{6}\right)=\frac{5}{12}$

P(loss in 1 round) $=\frac{5}{12}$

$$P(X_2>Y_2)=P(10,0)+P(7,2)+P(5,4)=\frac{5}{12}\times\frac{5}{12}+\frac{5}{12}\times\frac{1}{6}\times 2+\frac{5}{12}\times\frac{5}{12}\times 2=\frac{95}{144}$$

$$P(X_2=Y_2)=P(4,4)=\frac{1}{6}\times\frac{1}{6}=\frac{1}{36}$$

$$P(X_3=Y_3)=P(6,6)=\frac{1}{6}\times\frac{1}{6}\times\frac{1}{6}=\frac{1}{216}$$

$$P(X_3>Y_3)=P(15,0)+P(10,4)+P(12,2)+P(7,6)+P(9,4)$$

$$=\frac{5}{12}\times\frac{5}{12}\times\frac{5}{12}+\frac{5}{12}\times\frac{5}{12}\times\frac{5}{12}\times 3+\frac{5}{12}\times\frac{5}{12}\times\frac{1}{6}\times 3+\frac{5}{12}\times\frac{5}{12}\times\frac{1}{6}\times 6+\frac{5}{12}\times\frac{1}{6}\times\frac{1}{6}\times 3$$

$$=\frac{5}{12}\times\frac{5}{12}\times\frac{5}{6}\times 2+\frac{5}{12}\times\frac{5}{12}\times\frac{1}{6}\times 9+\frac{5}{12}\times\frac{2}{12}\times\frac{1}{6}\times 3=\frac{505}{864}$$

15.

16. A) $I=\int\frac{dx}{a^2-x^2}=\frac{x}{a^2-x^2}-\int\frac{2xdx}{\left(a^2-x^2\right)^2}x=\frac{x}{a^2-x^2}+2\int\frac{a^2-x^2-a^2}{(a^2-x^2)^2}dx$

$$=\frac{x}{a^2-x^2}+2I-2a^2\int\frac{dx}{(a^2-x^2)^2}\qquad=\frac{x}{2a^2(a^2-x^2)}+c$$

B) Put $x=a\sec\theta\Rightarrow dx=a\sec\theta\tan\theta d\theta$

$$\therefore I=\int\frac{a\sec\theta\tan\theta d\theta}{a^2\sec^2\theta.a\tan\theta}=\frac{1}{a^2}\sin\theta+c=\frac{\sqrt{x^2-a^2}}{a^2x}+c$$

C) Put $x=a\sin\theta\Rightarrow dx=a\cos\theta d\theta$

$$\therefore I=\int\frac{a\cos\theta a\cos\theta d\theta}{a^2\sin^2\theta}=\int\cot^2 d\theta=-\cot\theta-\theta+c$$

$$=-\frac{\sqrt{a^2-x^2}}{x}-\sin^{-1}\frac{x}{a}+c=-\frac{\sqrt{a^2-x^2}}{x}+\cos^{-1}\frac{x}{a}+c$$

D) $x=a\sec\theta\Rightarrow dx=a\sec\theta\tan\theta d\theta\ \therefore I=\int\frac{a\tan\theta a\sec\theta\tan\theta d\theta}{a\sec\theta}=a\int\tan^2\theta d\theta$

$$=a\tan\theta-a\theta+c\quad=\sqrt{x^2-a^2}-a\sec^{-1}\frac{x}{a}+c$$

17. $f(x)=a(x+1)^2+b\qquad f(1)=5; f(-1)=1$

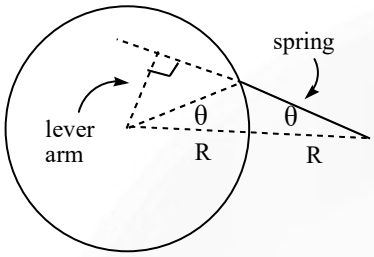
$f(x)=(x+1)^2+1\ =x^2+2x+2$ Now verify

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PHYSICS

18. **CONCEPTUAL**

19. Let θ be the angular displacement of the disk. For small the radius R. So the squat triangle in figure with base 2R is essentially isosceles. This implies that the angle on the right is equal to the angle on the left (which we defined to be θ), as shown. Second, the spring force has magnitude essentially equal to F = kR because the relaxed length is zero.



By looking at the thin right triangle with hypotenuse 2R in figure, the lever arm of the spring force, relative to the center of the disk, is $l=(2R)\sin\theta\approx 2R\theta$, where we have used the $\sin\theta\approx\theta$ small-angle approximation. The effect of the torque from the spring is to decrease θ. If θ is positive, so the torque is $\tau=-Fk=-(kR)(2R\theta)$.

The $\tau=I\alpha$ equation around the center of the disk is therefore

$$-(kR)(2R\theta)=\left(\frac{mR^2}{2}\right)\ddot{\theta}\Rightarrow\ddot{\theta}=-\frac{4k}{m}\theta\Rightarrow\omega=2\sqrt{\frac{k}{m}} \qquad .....(1)$$

20. $\frac{dU}{dN}=2KT$ (U is internal energy) $\qquad d\left[\frac{3}{2}NKT\right]=2KT\quad dN$

$$\frac{3}{2}NKdT+\frac{3}{2}KTdN=2KTdN \qquad 3\frac{dT}{T}=\frac{dN}{N};\ N=\frac{N_0}{8}$$

21. Conceptual

22. $\Delta p=-B\frac{\partial y}{\partial x}$

23. $W_{el}+W_{ext}=k_f-k_i \qquad qV_i-qV_f+W_{ext}=k_f-k_i$

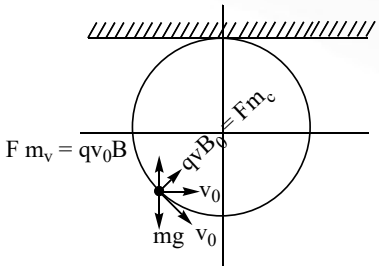
$$\frac{q\sigma}{2\varepsilon_0}\left(\sqrt{R^2+Z^2}-Z\right]-\frac{q\sigma R}{2\varepsilon_0}+CZ=k_f-0 \quad C=\frac{q\sigma B}{2\varepsilon_0}$$

Substitute β & Z, calculate kinetic energy at z = 0.

If kinetic energy is positive, then particle will reach at origin.

If kinetic energy is negative, then particle will not reach at origin.

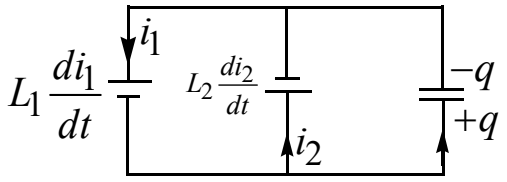
24.



$$(1)\,av_0B=\frac{3v_0^2}{R} \qquad (2)\,qv_0B=gm$$

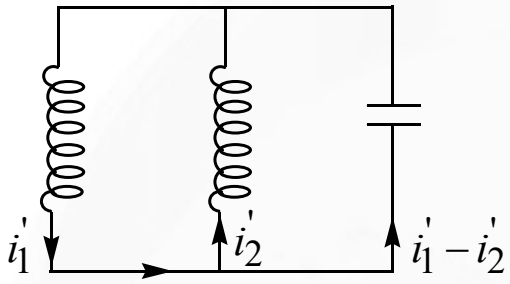
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25.



So, $L_1\frac{di_1}{dt}=-L_2\frac{di_2}{dt}=\frac{-q}{C}$

For maximum current in L_2:



$$\Rightarrow L_1\left(i'_1-\frac{E}{r}\right)=-L_2\left(i'_2-0\right) \qquad \frac{1}{2}L_1\left(\frac{E}{r}\right)^2=\frac{1}{2}L_1i'^2_1+\frac{1}{2}L_2i'^2_2$$

$$\Rightarrow \frac{L_1E^2}{r^2}=L_2i'^2_2+L_1\left(\frac{E}{r}-\frac{L_2i'^2_2}{L_1}\right)^2 \qquad \Rightarrow i'_2=\frac{2EL_1}{r(L_1+L_2)}$$

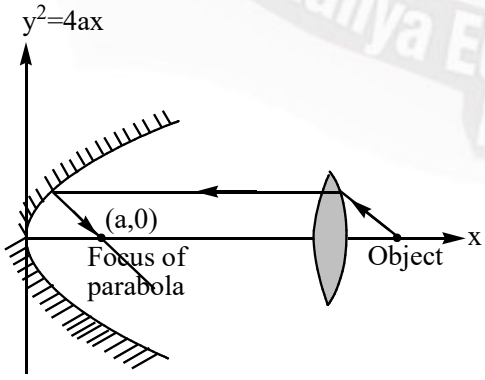
$$E=5V,\ L_1=4mH,\quad L_2=1mH,\ r=2\Omega \Rightarrow i'_2=\frac{2\times5\times4}{5\times2}=4A$$

26. $\lambda_{th}=\frac{1240}{2}=620\,nm \qquad i=\frac{dN}{dt}e=\frac{e\times A}{hc}\int\lambda dI=\frac{eAk\int_{400}^{620}\lambda d\lambda}{hc}$

$$=\frac{eAk\lambda^2}{2hc}\Big|_{400}^{620}=\frac{1\times1\times(620^2-400^2)\times k}{1240\times2} \qquad =\frac{220\times1020}{1240\times2}\times\frac{0.031}{100}=2.8A\approx3A$$

27. Comparing with $y^2=4ax\Rightarrow a=0.5$

$\Rightarrow$ final position of image = 0.5m = 50 cm



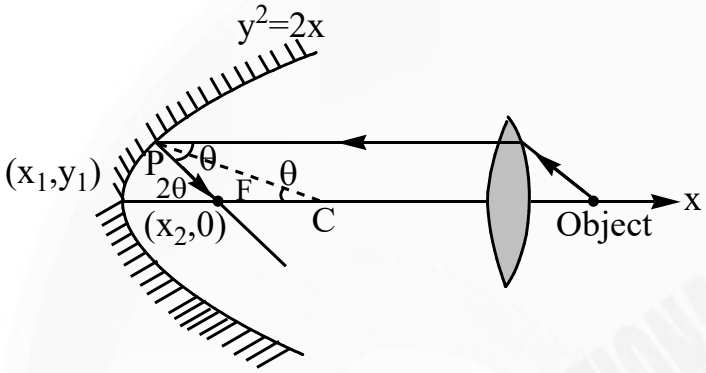
OR

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PC is a normal so $(\pi-\theta)=\dfrac{-1}{\left(\dfrac{dy}{dx}\right)_{x_1y_1}}=-y_1$

But $\tan 2\theta=\dfrac{y_1-0}{x_2-x_1}$ & $\tan 2\theta=\dfrac{2\tan\theta}{1-\tan^2\theta}$

$\Rightarrow \dfrac{y_1}{x_2-x_1}=\dfrac{2(y_1)}{1-y_1^2}\Rightarrow x_2=\dfrac{1}{2}m=0.5m=50cm$



28. Path difference produced by glass slab $=\left(1-\dfrac{\mu_2}{\mu_1}\right)t$

$\mu_2\rightarrow$ R.I. of film $\mu_1\rightarrow$ R.I. of medium

Now, $\Delta x=\left(1-\dfrac{\mu_2}{\mu_1}\right).t=\dfrac{n.\lambda}{\mu}\qquad\Rightarrow\left(1-\dfrac{1}{3/2}\right)t=\dfrac{n.\lambda}{\mu}$

$\Rightarrow\left(1-\dfrac{2}{3}\right)t=\dfrac{n.\lambda.2}{3}\Rightarrow\dfrac{t}{3}=\dfrac{2n\lambda}{3}\Rightarrow n=\dfrac{1}{2\lambda}=\dfrac{4\times10^{-6}}{2\times4000\times10^{-7}}\qquad=\dfrac{4\times10}{2\times4}=\dfrac{40}{8}=5$

29. $E_0=200\qquad I=\dfrac{1}{2}\varepsilon_0E_0^2.c$

Radiation pressure

$P=\dfrac{2I}{c}=\left(\dfrac{2}{c}\right)\left(\dfrac{1}{2}e_0E_0^2c\right)=\varepsilon_0E_0^2$

$=8.85\times10^{-12}\times200^2\qquad=8.85\times10^{-8}\times4=\dfrac{354}{10^9}=354.0$

30. For stability pressure at the core (r = 0) must be balanced by the radiation pressure

i.e $\rho(r=0)=\dfrac{4\sigma}{3c}T_c^4\qquad\dfrac{5}{36}\pi G\alpha^2R^2=\dfrac{4\sigma}{3c}T_c^4$

$T_c=\left[\dfrac{5}{48}\dfrac{\pi G\alpha^2R^2c}{\sigma}\right]^{1/4}\qquad x=48$

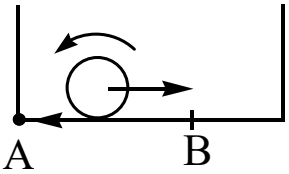
31. CONCEPTUAL

32. $e=-\dfrac{d\phi}{dt}$ & $\vec{\tau}=\vec{\mu}\times\vec{B}$

33. a) After collision $\upsilon\rightarrow$ but ω same ball cannot stop because

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$L_A = constant \neq 0$



The $\tau_{fr} = c\omega.L_B = const.$

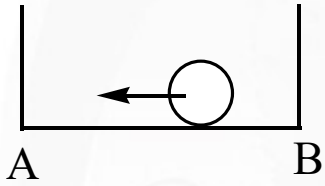
$(\tau_{fr} = 0), L_B = m\upsilon R - I\omega(c\omega)$

$= m\upsilon R - \frac{2}{5}mR^2 \times \frac{\upsilon}{R} = \frac{3}{5}m\upsilon R(c\omega)$

$\Rightarrow$ If friction is sufficient, ball may rotate $c\omega$ or not rotate at all. Energy is dissipated due to friction.

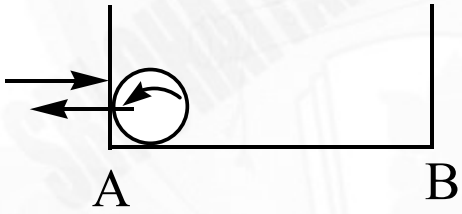
b) No friction $\Rightarrow$ cannot stop

$L_B =$ constant no change in motion.



c) Ball cannot stop because friction is non-impulsive. Friction oppose its rotation $\tau_N \neq 0$.

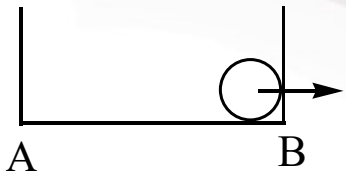
$\Rightarrow L_B \pm$ constant τ_{fr} is small $\Rightarrow \omega$ does not change, mechanical energy decreases slightly due to friction.



d) Ball may be rotating $c\omega$, a $c\omega$ or not at all. If cannot stop (elastic) friction may oppose rotation if rotating.

L_B is not constant due to τ of f_r. And N. If rotating $ac\omega$, $\tau_{fr} = c\omega$.

$\int N\,dt = 2m\upsilon_1$



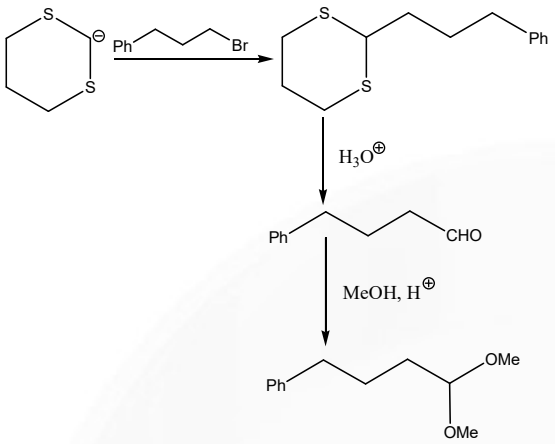
τ of $\mu\int N$ may make it rotate $c\omega$. Due to friction, energy may decrease.

34. CONCEPTUAL

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CHEMISTRY

35.



36. $P_A = P_B \qquad XA.P_A^0 = X_B.P_B^0$

$$\frac{X_A}{X_B} = \frac{P_B^0}{P_A^0} \qquad 1+\frac{X_A}{X_B} = 1+\frac{P_B^0}{P_A^0}$$

$$\frac{1}{X_B} = \frac{P_A^0 + P_B^0}{P_A^0} \qquad X_B = \frac{P_A^0}{P_A^0 + P_B^0}$$

Similarly, $X_A = \dfrac{P_B^0}{P_A^0 + P_B^0}$ $P_T = P_A^0.X_A + P_B^0.X_B = \dfrac{2P_A^0.P_B^0}{P_A^0 + P_B^0}$

37. $Zn + 2H^+ \longrightarrow Zn^{+2} + H_2$

$$E_{cell}^0 = E_{ox}^0\left(Zn / Zn^{+2}\right) + E_{Rd}^0\left(H^+ / H_2\right) = 0.76 Volt$$

$$E_{cell} = E_{cell}^0 - \frac{0.0591}{2}\log\frac{\left[Z_n^{+2}\right].P_{H_2}}{\left[H^+\right]^2}$$

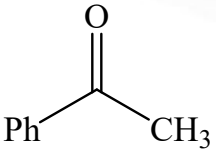
$$\left(H^+\right) = 10^{-2} M \qquad \left[OH^-\right] needed = 10^{-2} M$$

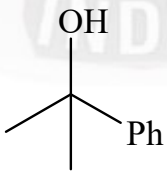
Wt of $NaOH$ needed = 0.4 gram

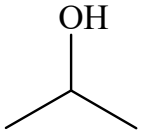
38. $XeO_3 + NaOH \rightarrow Na\left[H\ XeO_4\right]$

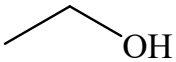
$$Na\left[H\ XeO_4\right] + NaOH \rightarrow Na_4\left[XeO_6\right] + Xe + O_2 + H_2O$$

39.

A = 

B = 

C = 

D = 

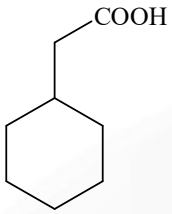
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40.

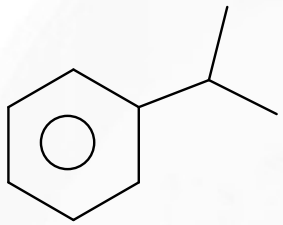
41. chloral hydrate: $Cl_3CCH(OH)_2$

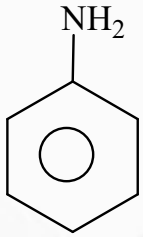
Hydrogen bonding: 5 membered ring

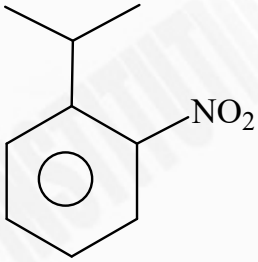
42.

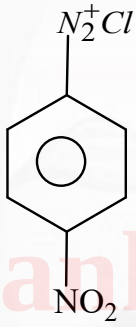
Major product = 

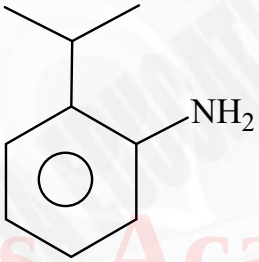
43.

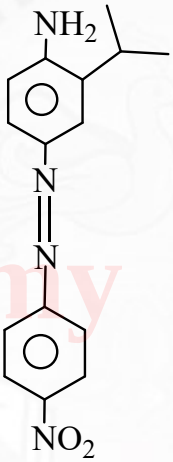
P = 

Q = 

R = 

S = 

T = 

U = 

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44. For the isothermal reversible expansion of 3.0 moles of an ideal monatomic gas from 4L to 8L at 400K.

$$\Delta S_1 = nR\ln\frac{v_2}{v_1} = 3\times 8.314\times \ln 2 - 17.46\ J/K$$

For isochoric cooling until pressure is 1/8 of initial.

$$\Delta S_2 = nC_{v,m}\ln\frac{T_2}{T_1} = 3\times\frac{3}{2}R\times\ln\frac{100}{400} = -52.38\ J/K$$

Total $\Delta S = \Delta S_1 + \Delta S_2 = -34.92\ J/K$

$|\Delta S| = 35\ J/K$

45. $$E^0_{cell} = +\frac{0.06}{1}\log\frac{10^{-10}}{10^{-15}}$$

$E^0_{cell} = +0.3V$

$\therefore$ Energy incident $= 0.3+5.4 = 5.7eV$

$Now incident\ energy = 2\times 5.7 = 11.4$

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$\therefore K.E_{max} = 11.4 - 5.4 = 6.0eV$

$\lambda \approx \sqrt{\frac{150}{V}} = 5\overset{0}{A}$

46. $HClO_4 : Cl = +7$ (Oxidation state)

$F_2 + NaOH \rightarrow OF_2 + NaF + H_2O$

$F_2 + NaOH \rightarrow O_2 + NaF + H_2O$

$X_2 + NaOH \rightarrow NaX + NaOX / NaXO_3 + H_2O$

$(X = Cl, Br, I)$

$Cl_2O + NaOH \rightarrow NaOCl + H_2O$

$Cl_2O_7 + NaOH \rightarrow NaClO_4 + H_2O$

47. 12th vertical column: Zn family

3rd element: Hg $(Z_1 = 80)$

17th vertical column : Halogens

3rd element: Br $(Z_2 = 35)$

13th vertical column: boron family

3rd element: Ga $(Z_3 = 31)$

48. P= SN^2 reaction

Q = strecker synthesis

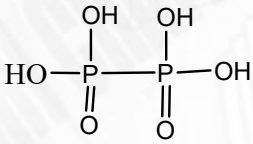
R = gabriel synthesis

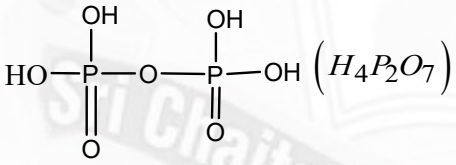
D = nucleophilic edition

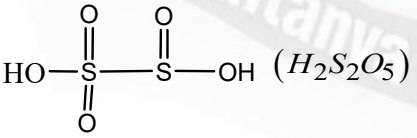
49. P gives Diastereoisomers, Q gives meso, R = racemic mixture, S gives meso

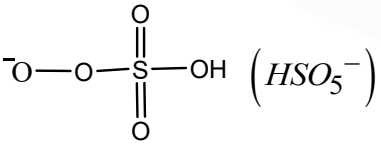
50. $A-q,s,t : B-p : C-q : D-r$

51.

P)  $(H_4P_2O_6)$

Q)  $(H_4P_2O_7)$

R)  $(H_2S_2O_5)$

S)  (HSO_5^-)

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